WEEK 16 · Generative Monte Carlo

Monte Carlo and Generative AI

What risks appear when we generate many possible markets?

Monte Carlo · Block Bootstrap · GAN · Diffusion · Joint Paths · Capstone

Today's question — if we generate many future paths and apply the same operating rules, how can we judge loss tails, cash depletion, and strategy fragility?

Change the generator, keep the financial evaluation the same

Three messages running through this lecture

1

Monte Carlo is a computation method, not an engine

Generate paths repeatedly, apply the same rules, and summarize P&L. The generator is a replaceable part.

2

Generative models are candidates to enrich joint paths

GANs and Diffusion can learn multi-asset co-movement and tails, but do not guarantee crisis correlations.

3

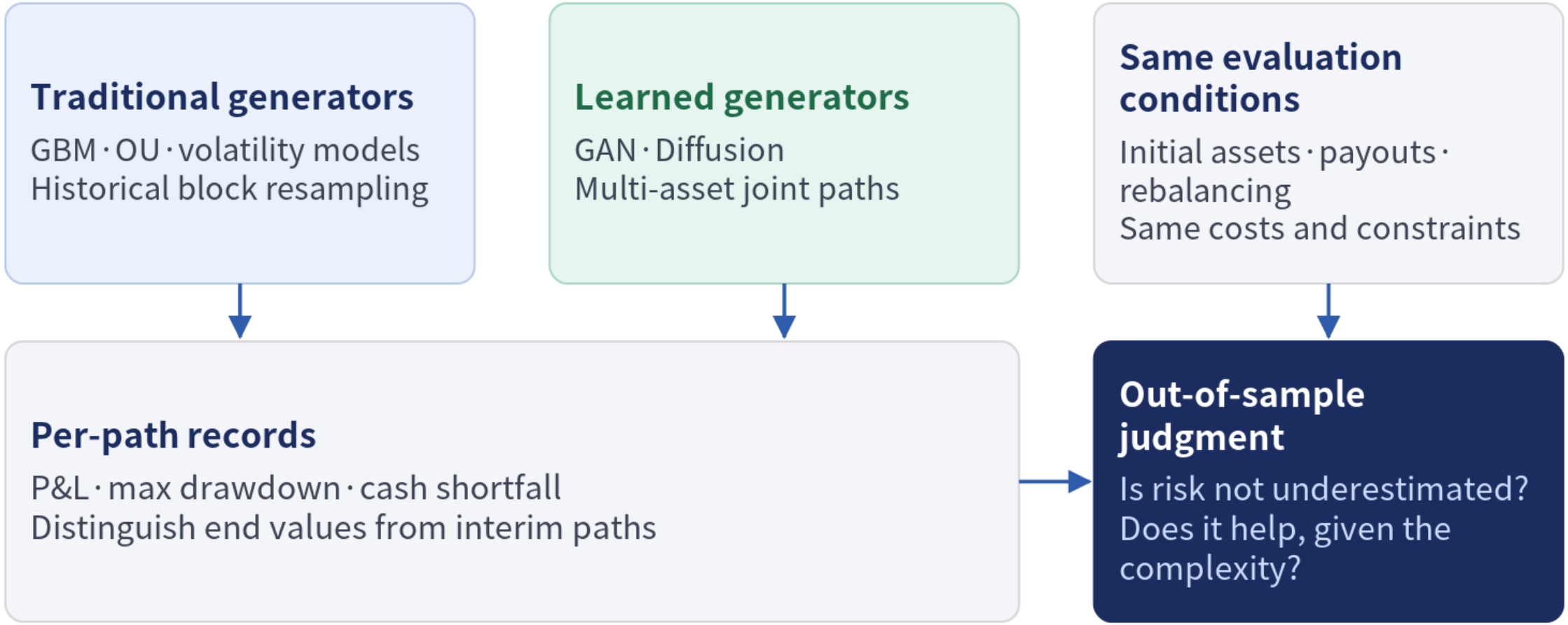
Adoption is judged by out-of-sample risk forecasting

The criterion is not pretty paths but whether tails, joint losses, and operating results improve in the real future.

Generating more paths does not add market information

Design the generator and the evaluation conditions separately

Connect traditional and learned generators to the same evaluation conditions, and judge per-path records out of sample



01

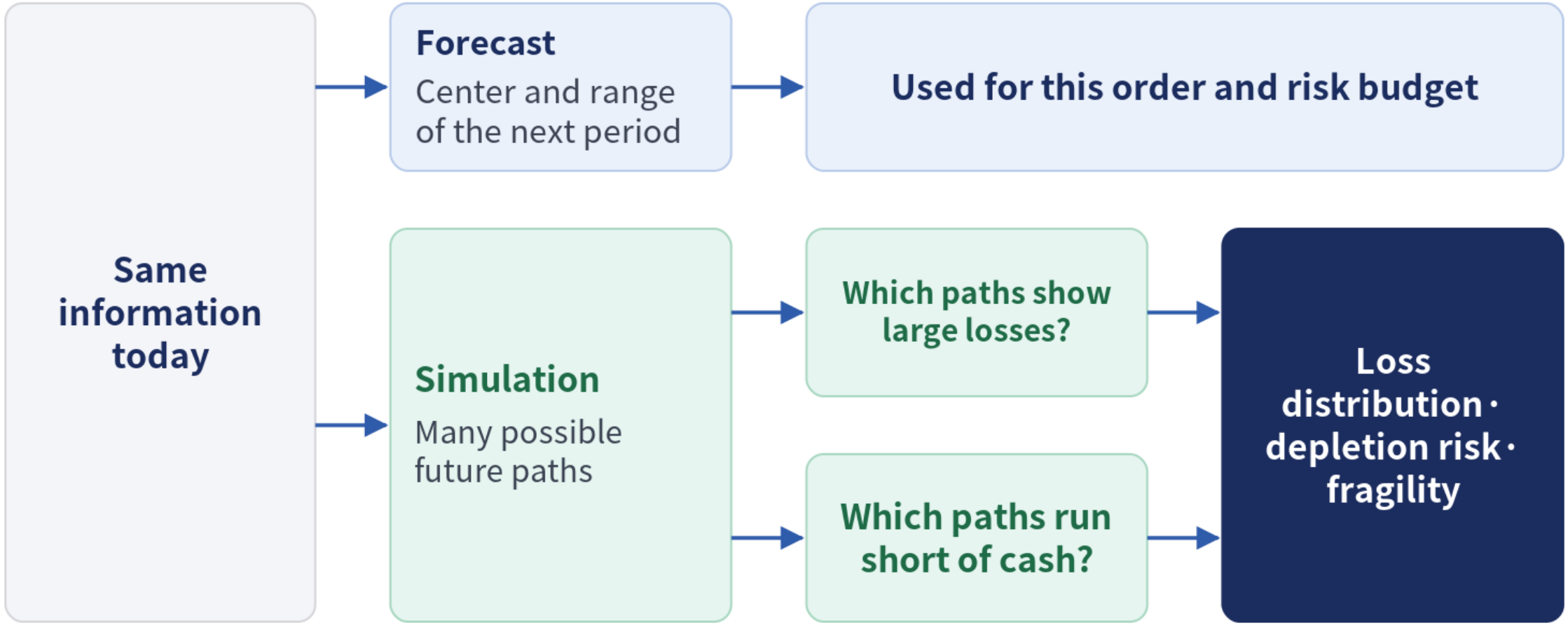
PART 1 · Monte Carlo Basics

Possible Future Paths and Loss Distributions

Forecasting vs. simulation · Traditional generators · KRW 10bn portfolio · Tails

Forecasting and simulation ask different questions

Forecasting asks for next period's center and range; simulation asks about losses and cash shortfalls across many paths



Several quantiles are not a joint path

Conditional generative models can link forecasting and simulation

Consistent multi-asset, multi-period joint paths cannot be built by stitching quantile forecasts together

What forecasting answers

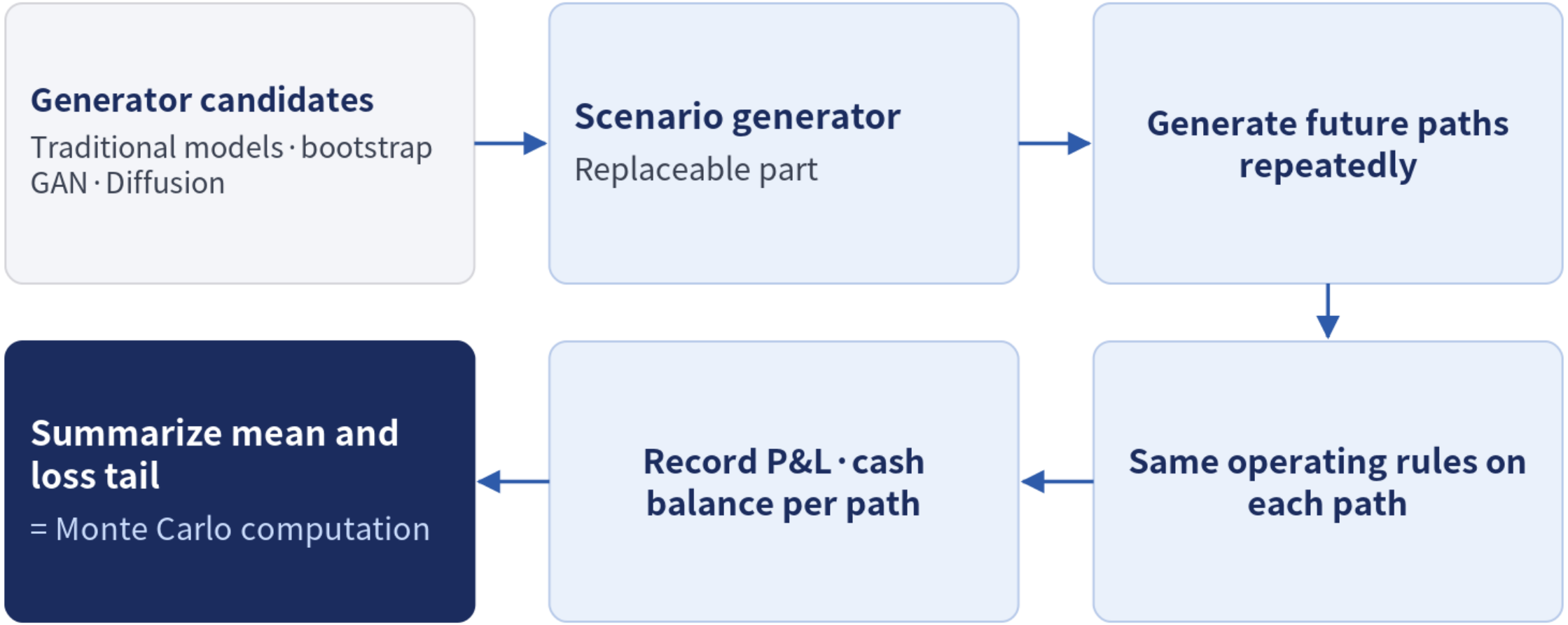
- Median and range for the next period
- This order and risk-budget decision

What simulation answers

- Which paths produce large losses?
- Which paths run short of cash?

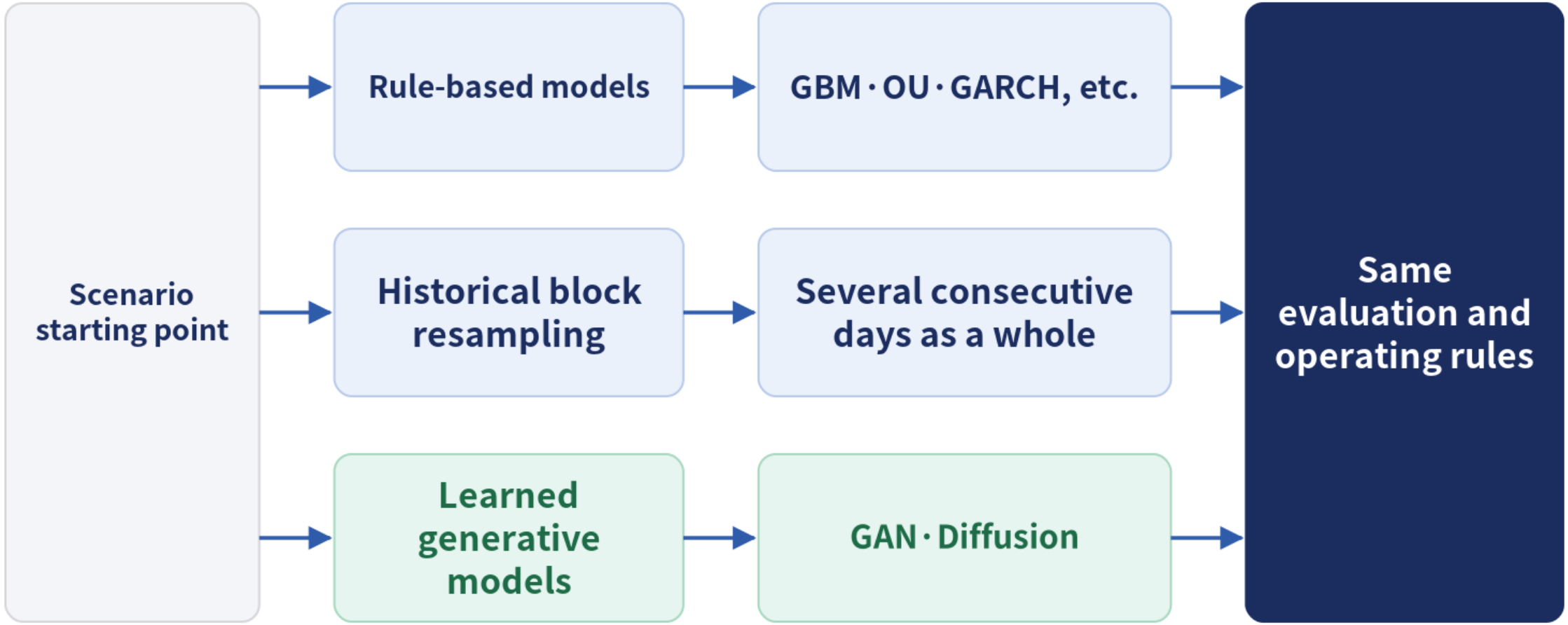
Monte Carlo is a computation method, not an engine

Even if the generator is a GAN, repeatedly generating paths and summarizing losses is still Monte Carlo



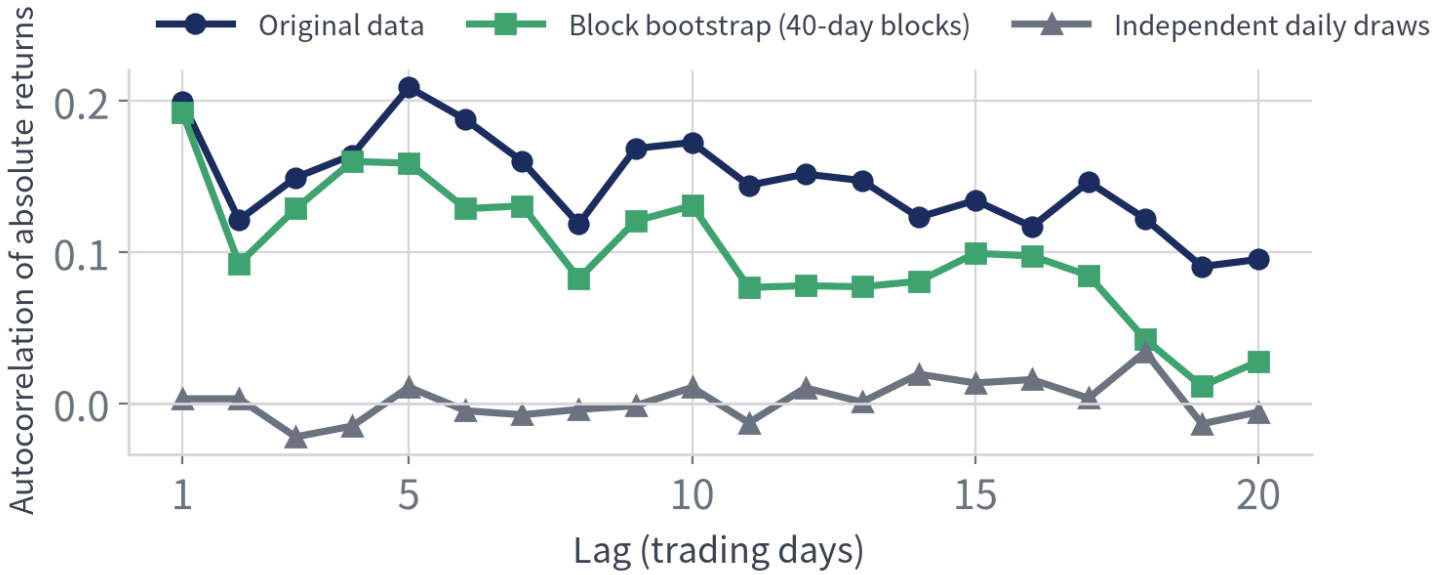
Scenario generators fall into three families

Rule-based models · Historical block resampling · Learned generative models — evaluation and operating rules stay the same



Drawing single days independently destroys volatility clustering

Autocorrelation of absolute returns for GARCH-type synthetic returns · ※ Synthetic data



Multiple Assets

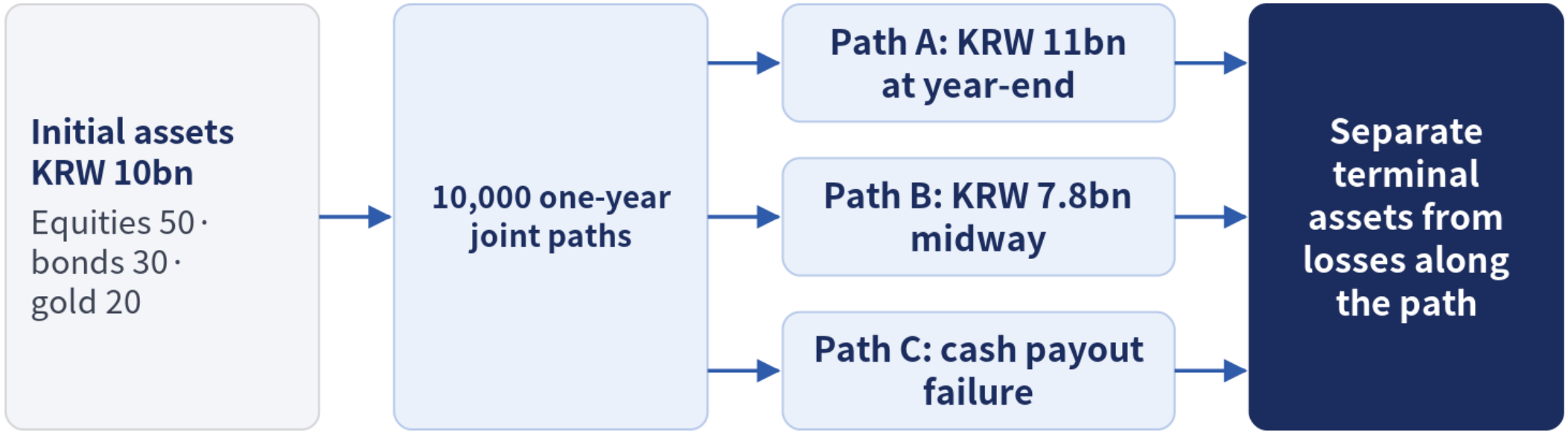
Drawing same-date blocks together preserves co-movement

Block Length

Length must also be validated; distortion at block joins remains

View one year of a KRW 10bn portfolio as paths

KRW 11bn and 7.8bn are illustrative path values, not computed results



Example conditions (teaching assumptions)

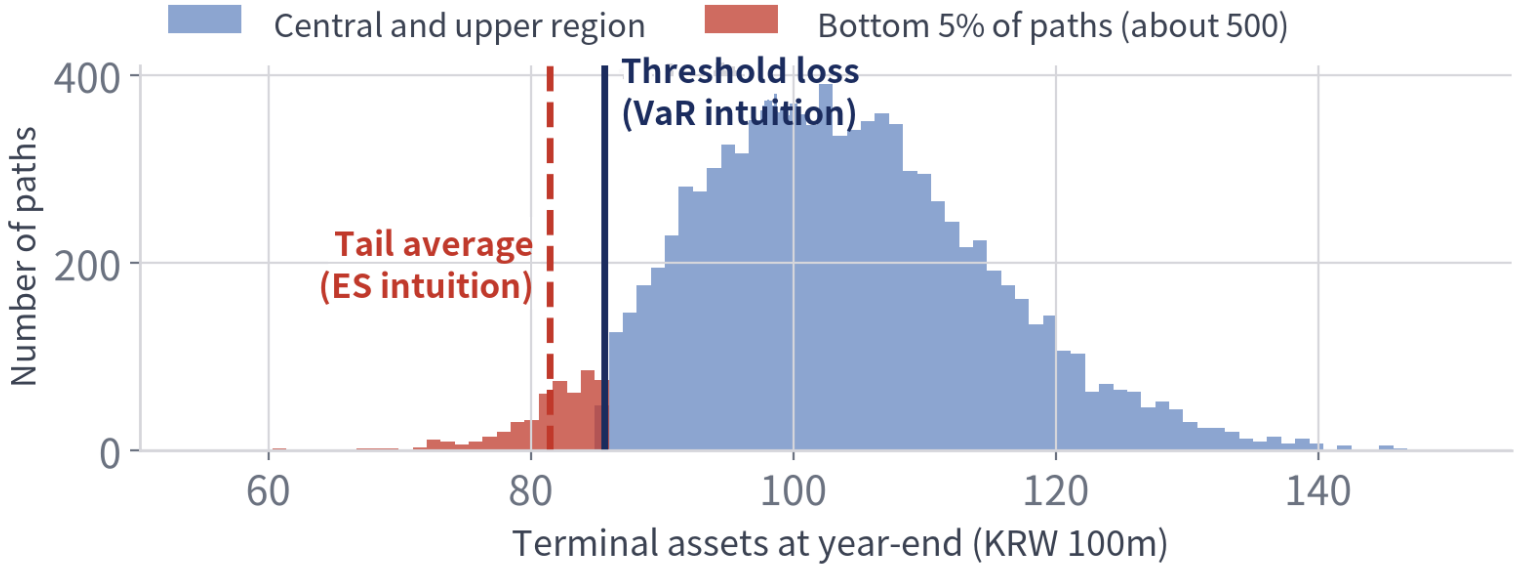
KRW 50m paid out monthly · Quarterly rebalancing to target weights · 10bp transaction cost

Rules to fix first

Cover shortfalls by selling, or keep a cash buffer?

Looking only at the mean misses the lower tail

Sort terminal wealth across paths and look at the bottom 5% · ※ Illustrative distribution (not a computed result)



Risk Along the Path

Maximum drawdown · Frequency of paths with insufficient balance on payout dates

Tail Sample Size

The bottom 5% of 10,000 paths is about 500 paths; the 99.9% tail is only about 10

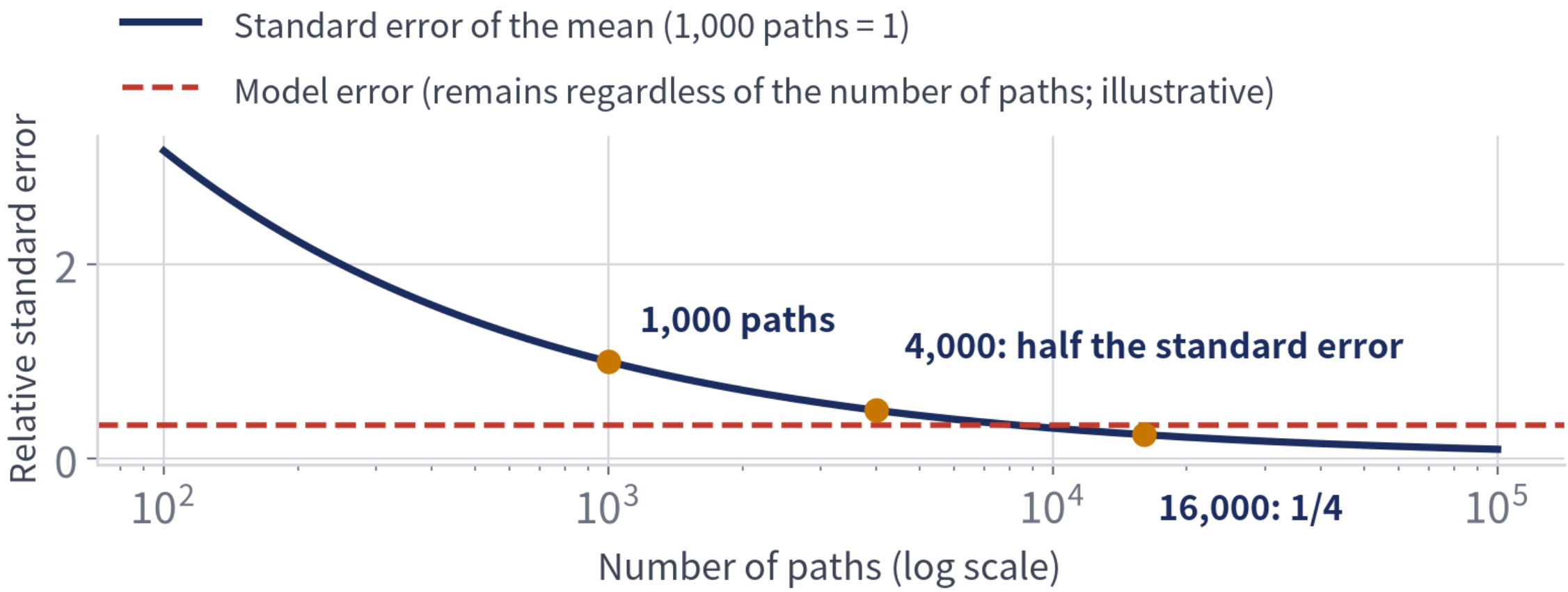
State confidence level, horizon, and sign together with risk metrics

Frequencies among generated paths are frequencies under that generative model, not confirmed real-world probabilities

Metric	Intuition	Specify together
VaR	Threshold loss where the lower tail begins	Confidence level · horizon · loss sign
ES	Average loss over the whole tail	Number of paths in the tail
Maximum Drawdown	Largest peak-to-trough decline along the path	Based on the path, not just the end value
Cash Depletion Frequency	Share of paths with insufficient balance on payout dates	Payout and selling rules

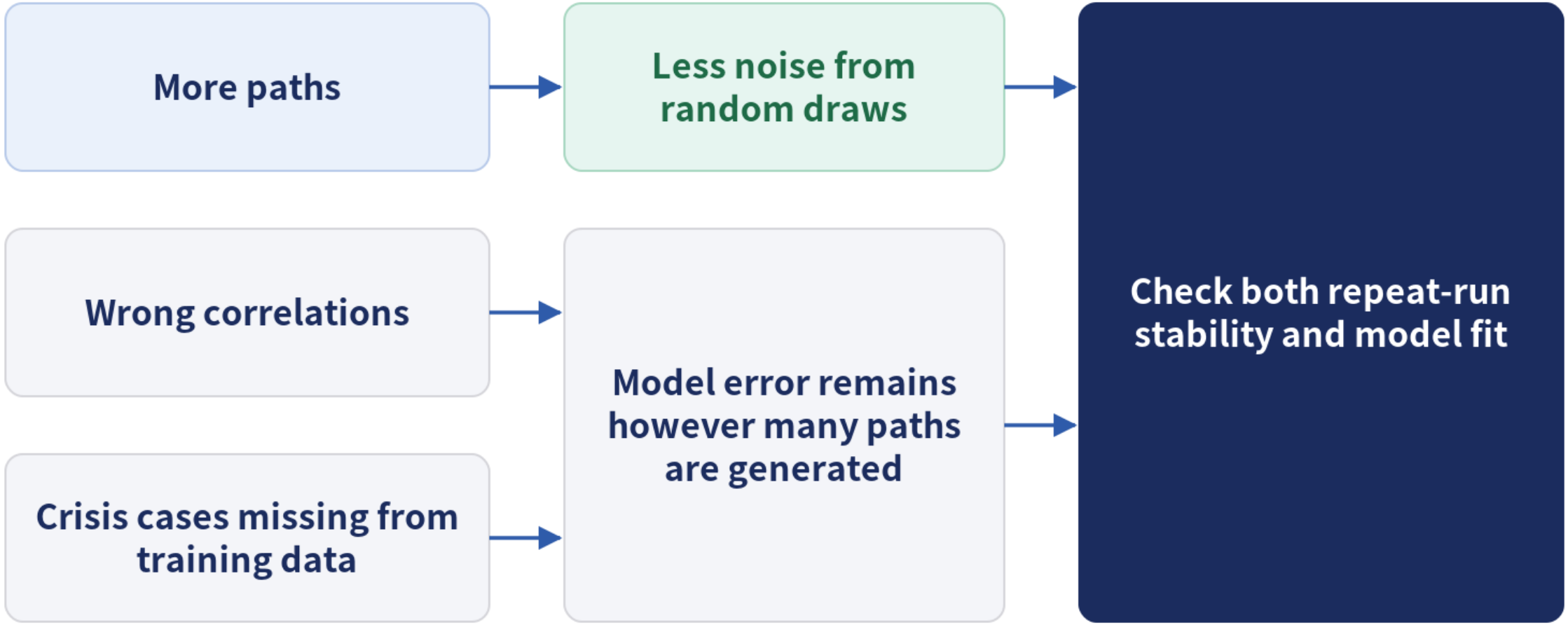
Quadrupling the paths halves the standard error

With independent random paths, the standard error of the mean is inversely proportional to the square root of the number of paths · Red line is illustrative



More paths still leave model error

Generating more paths does not add real market information



02

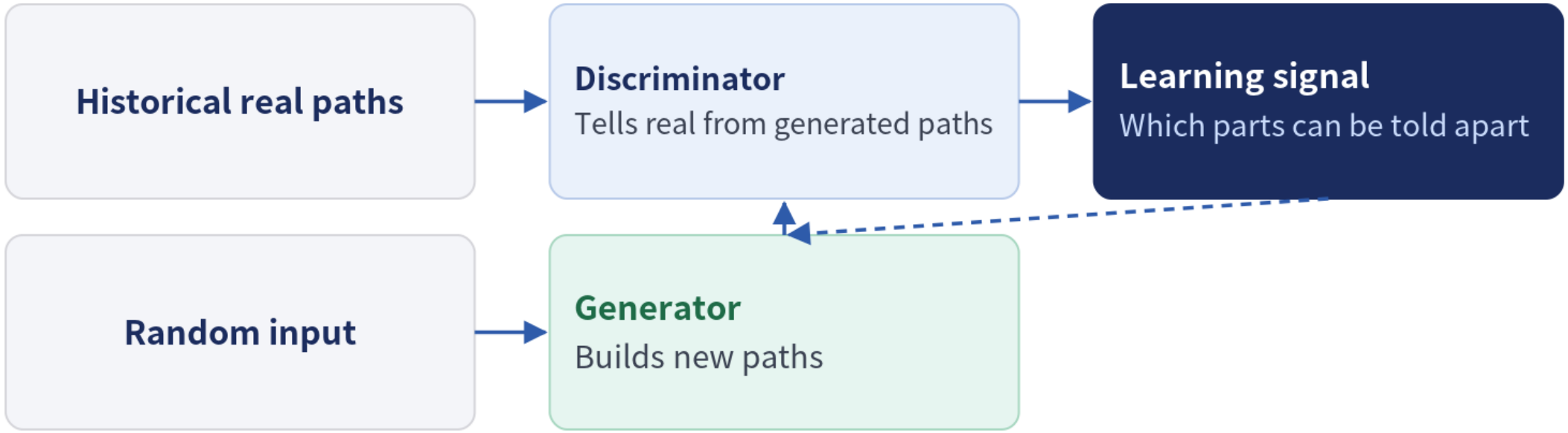
PART 2 · Enriching the Generator

Enriching with GANs and Diffusion

GAN · TimeGAN · Quant GAN · Diffusion · Conditional Generation · Hybrid Approaches

A GAN trains a generator and a discriminator in turns

The discriminator does not understand the economy — it is only trained to tell training data from generated data



Mode Collapse

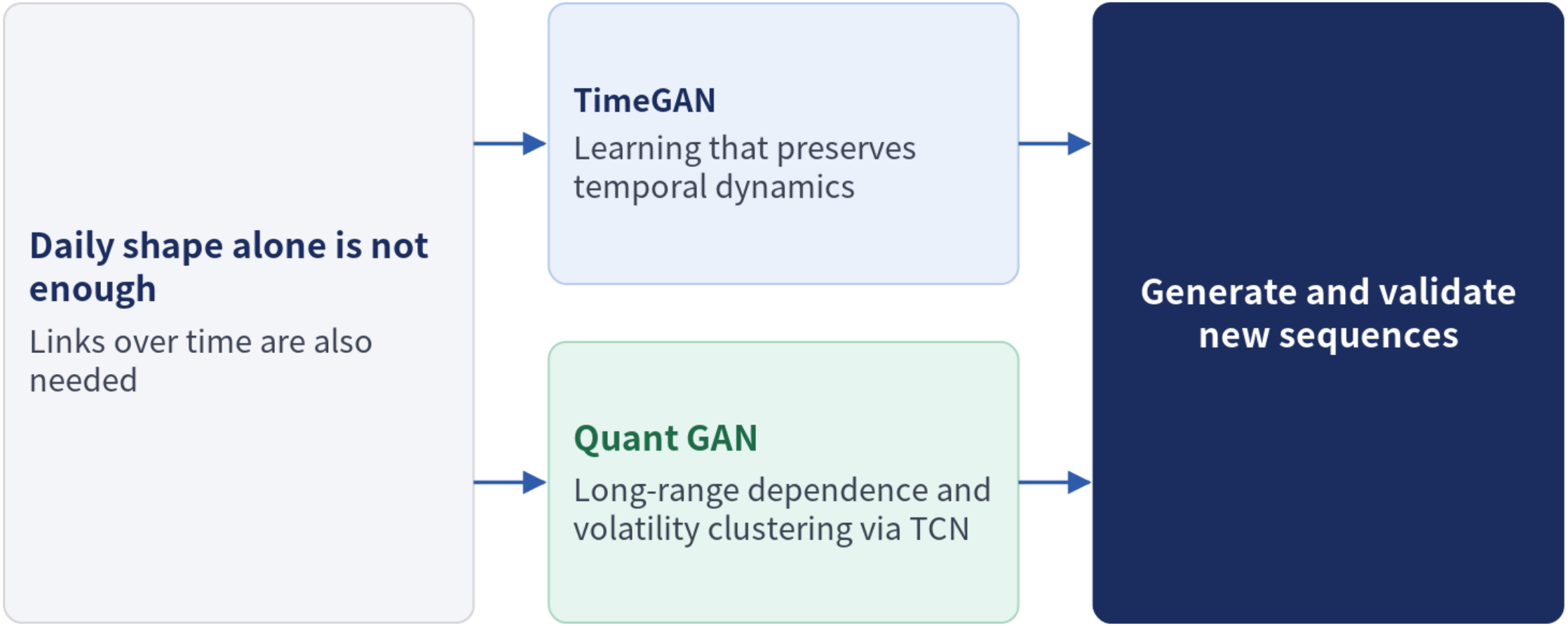
Generates only overly similar paths

Missing Rare Crashes

May fail to reproduce crises that are rare in the data
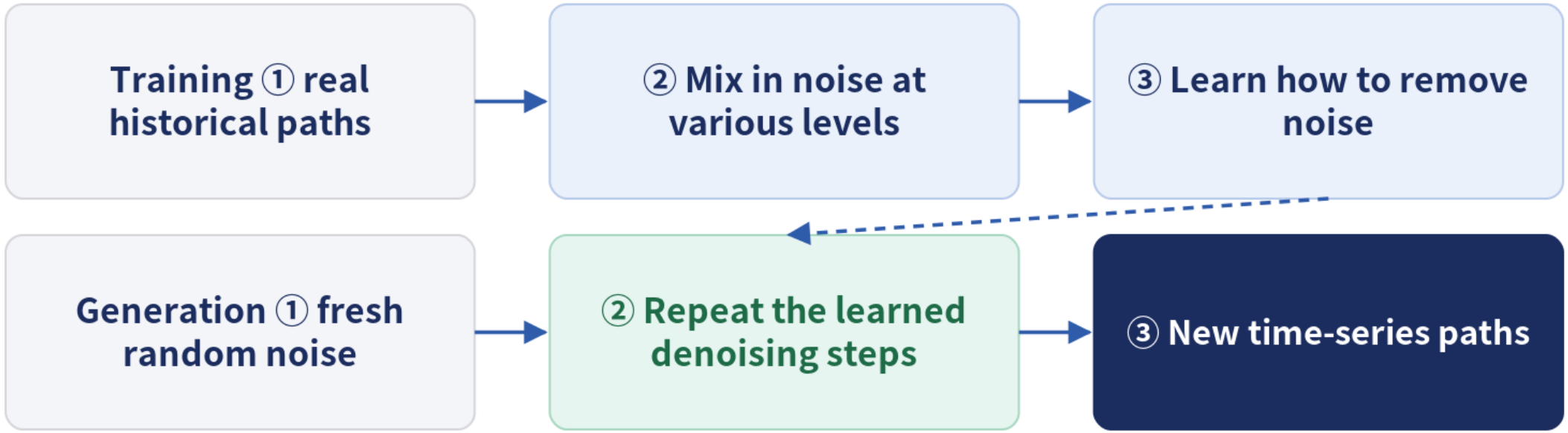
Time-series generation must also match temporal dependence

TimeGAN and Quant GAN illustrate basic generative concepts rather than a race for the latest version [S5][S6]



Diffusion trains and generates in opposite directions

Denoising steps are an artificial generation order, not the order in which market dates flow [S7]



Training

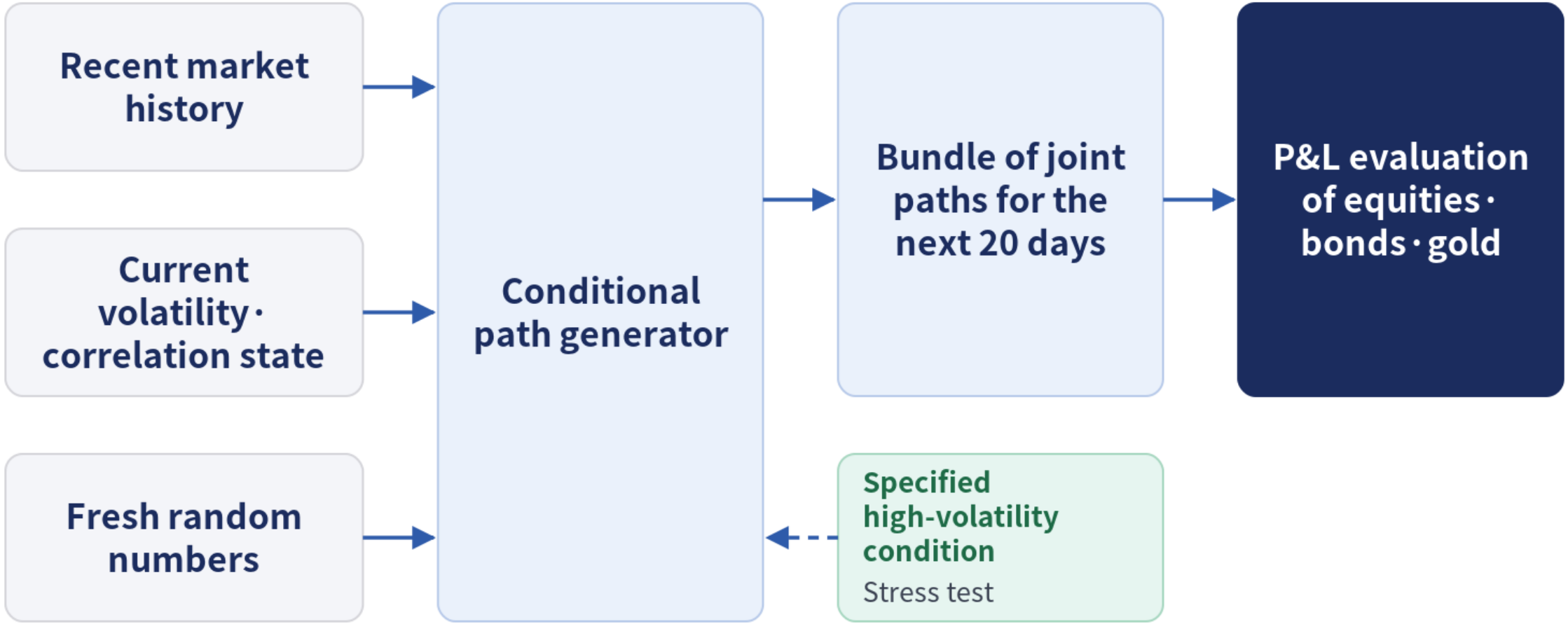
Learns to add noise to real paths and remove it

Generation

Starts from fresh noise and builds new paths consistent with the learned distribution

Conditional generation builds futures that reflect today's situation

TimeGrad is a leading example of multivariate probabilistic forecasting conditioned on the past [S8]



Distinguish observed conditions from assumed stress

Feeding in an arbitrary condition does not reveal the real-world probability of a crisis

Conditional generation learned from observed conditions ≠ stress test that deliberately assumes a crisis

Conditional Generation

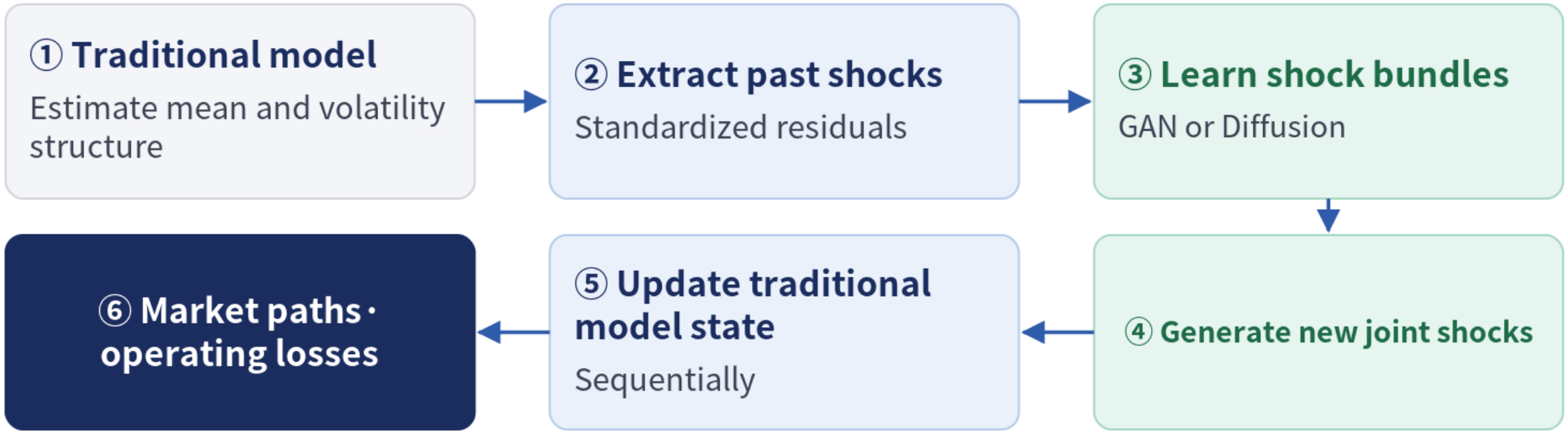
- Uses market history up to the forecast date and current volatility and correlation states as conditions

Stress Test

- Specify the high-volatility condition and state how future condition paths were assumed

Keep the traditional model and learn only the shocks

Leave volatility structure to the baseline model and use generated joint residuals instead of normal random draws



Checks

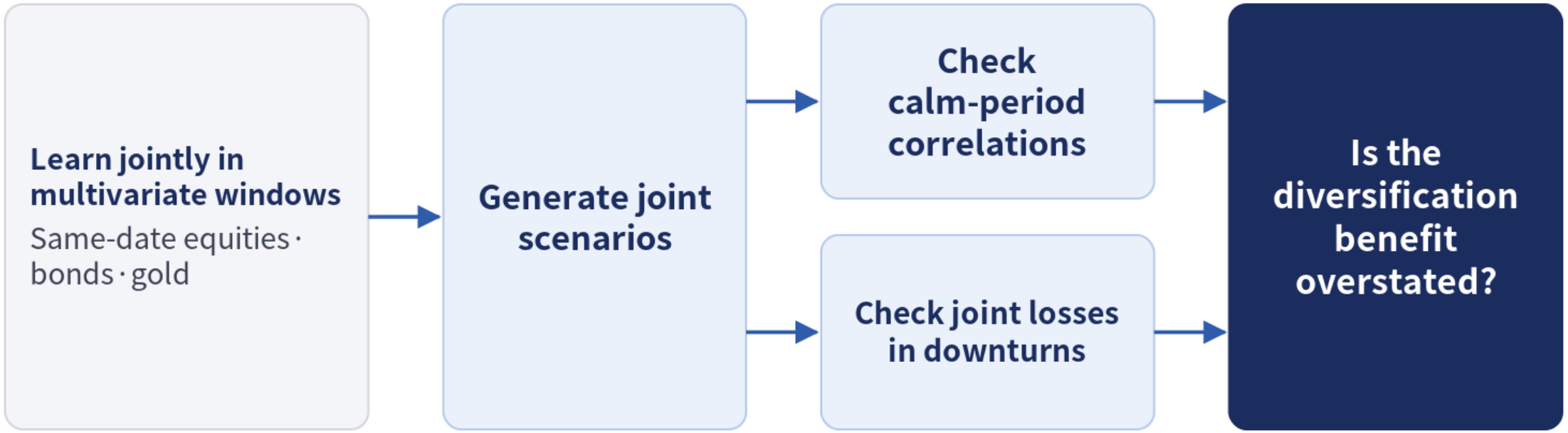
Units · mean · covariance

No Double Counting

Do not put volatility clustering already explained by the model back into the residuals

Generate multi-asset paths jointly

Generating each asset separately and then aligning dates does not preserve learned joint relationships

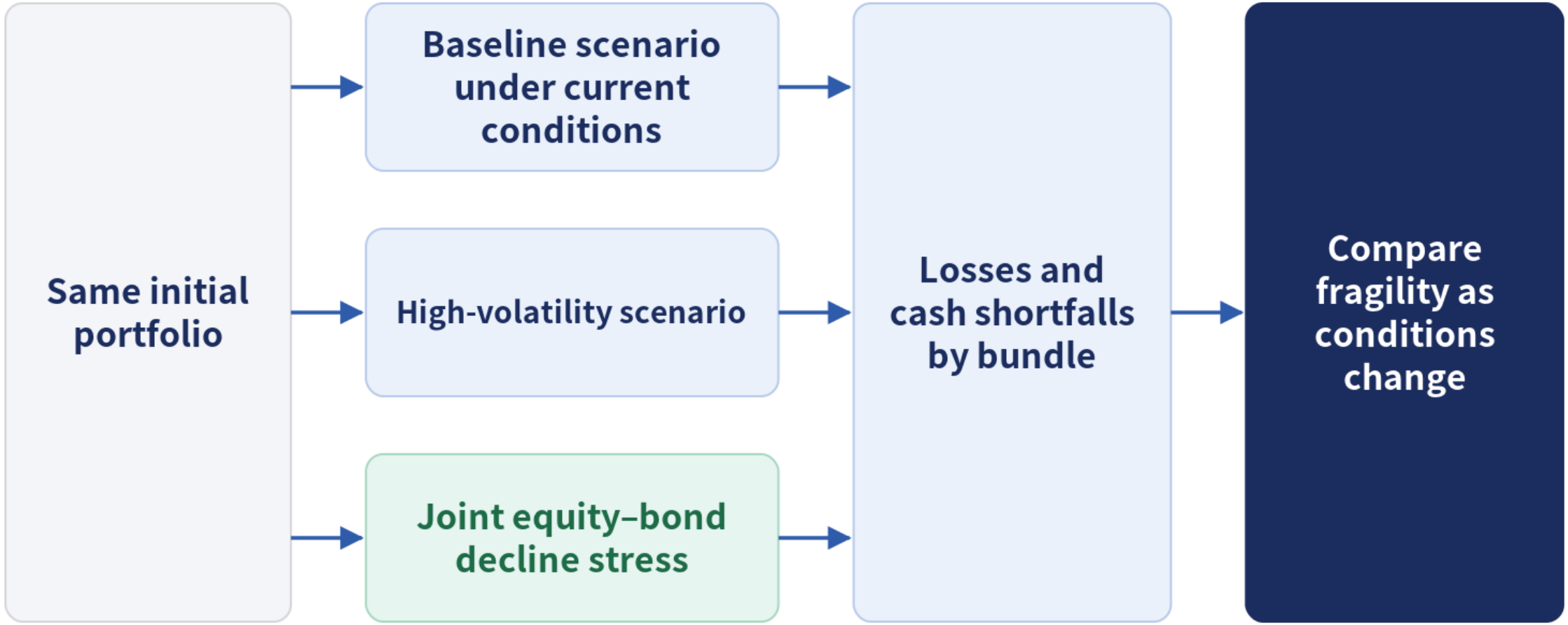


Limits

Matching historical correlations well does not guarantee crisis correlations or unseen policy regimes

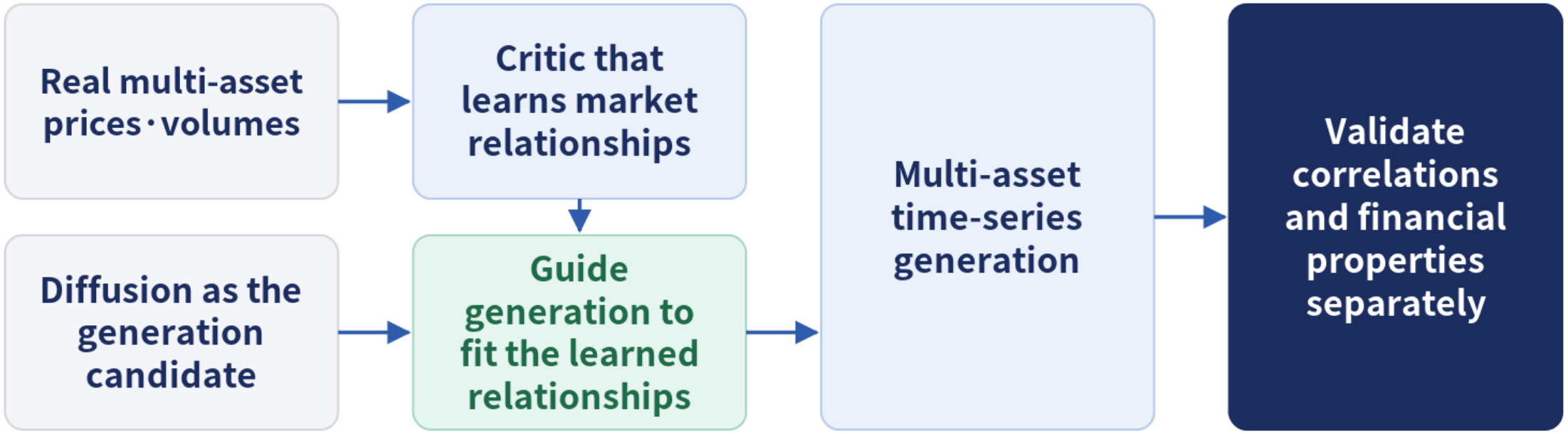
Compare scenarios separately by condition

Reporting baseline, high-volatility, and stress results mixed as a 'probability of occurrence' requires a basis for the condition weights



Studies combining GANs and Diffusion are emerging

2026 CoMeTS-GAN and GAN–Diffusion preprints — not evidence of general superiority or completed operational validation [S9]



Distinguish

Keeping only good-looking paths differs from guiding the generation process itself

03

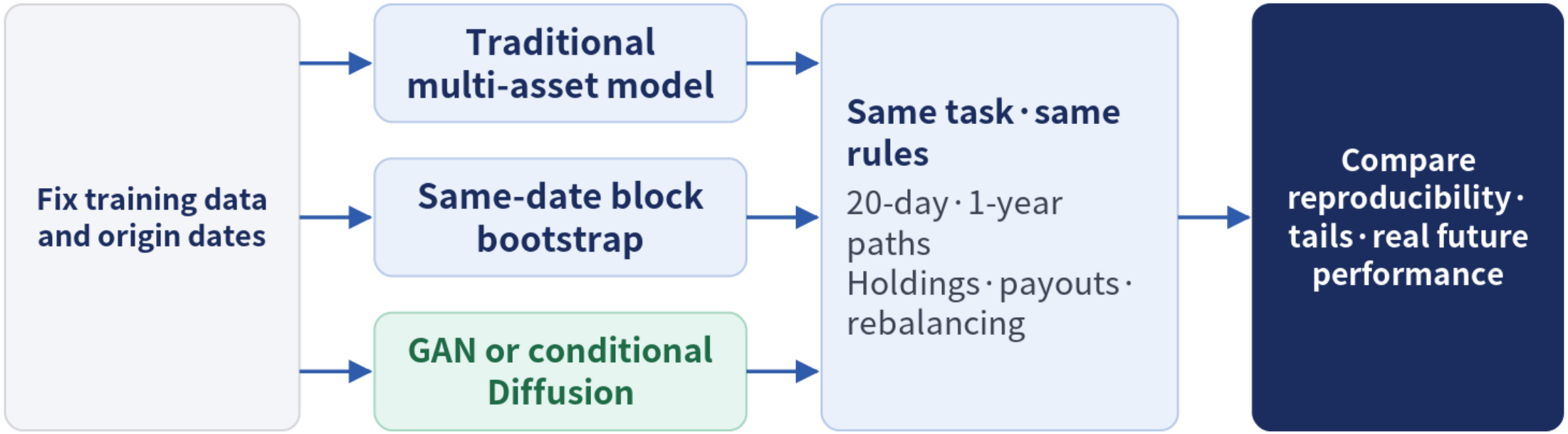
PART 3 · Judging the Enrichment

Judging Enrichment with Financial Cases

Same-condition comparison · Synthetic path validation · Out-of-time risk forecasting · Option pricing

Attach different generators to the same portfolio

Validate first with 20-day generation; check long-run consistency separately for long paths such as one year

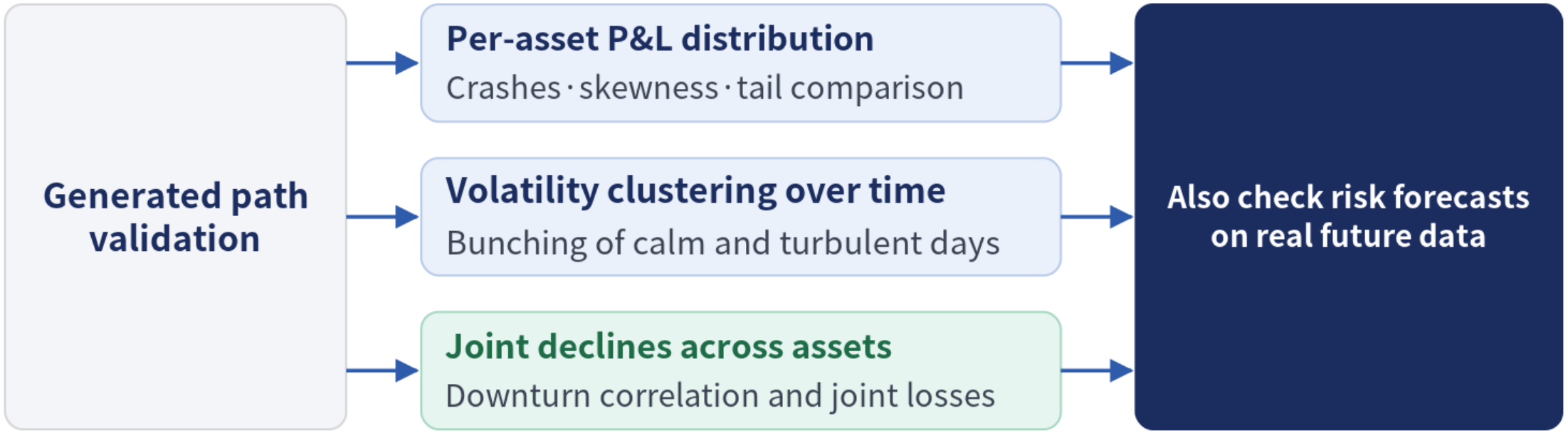


Experiment Design Principle

Do not present results of an untrained model as if it had been run — separate executed models from literature discussion

Synthetic time series must match on three things

Do not judge financial fitness from pretty path plots or a 'fooled the real/fake classifier' score alone

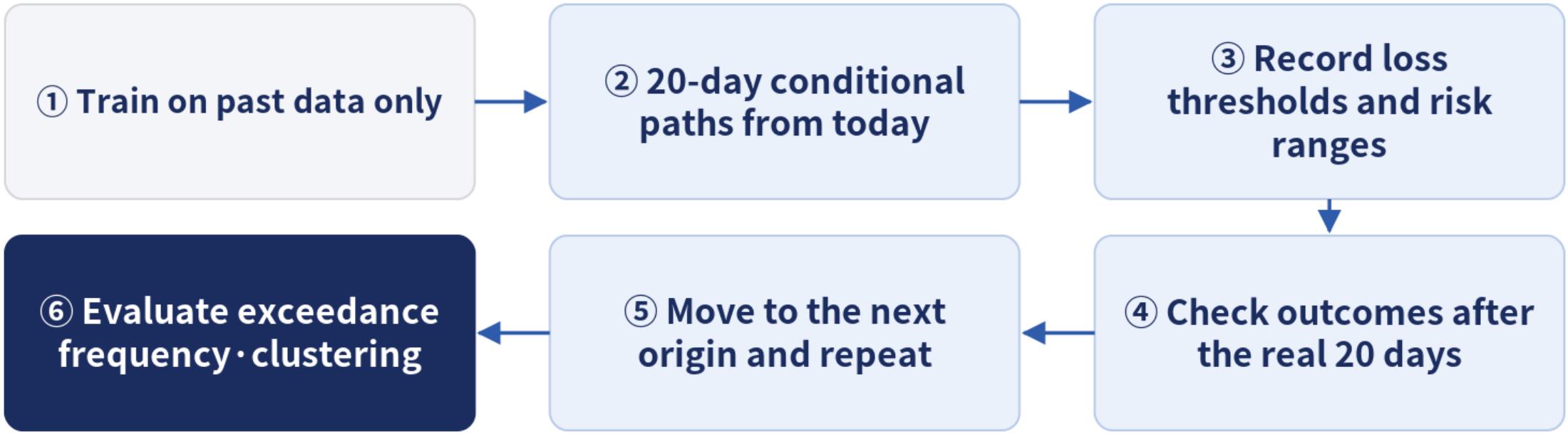


Copy Check

Check nearest paths and overlap to see whether training paths were nearly copied

Validate risk forecasts against the real future

Reproducing the ES of the training data well is not the same as forecasting future risk well

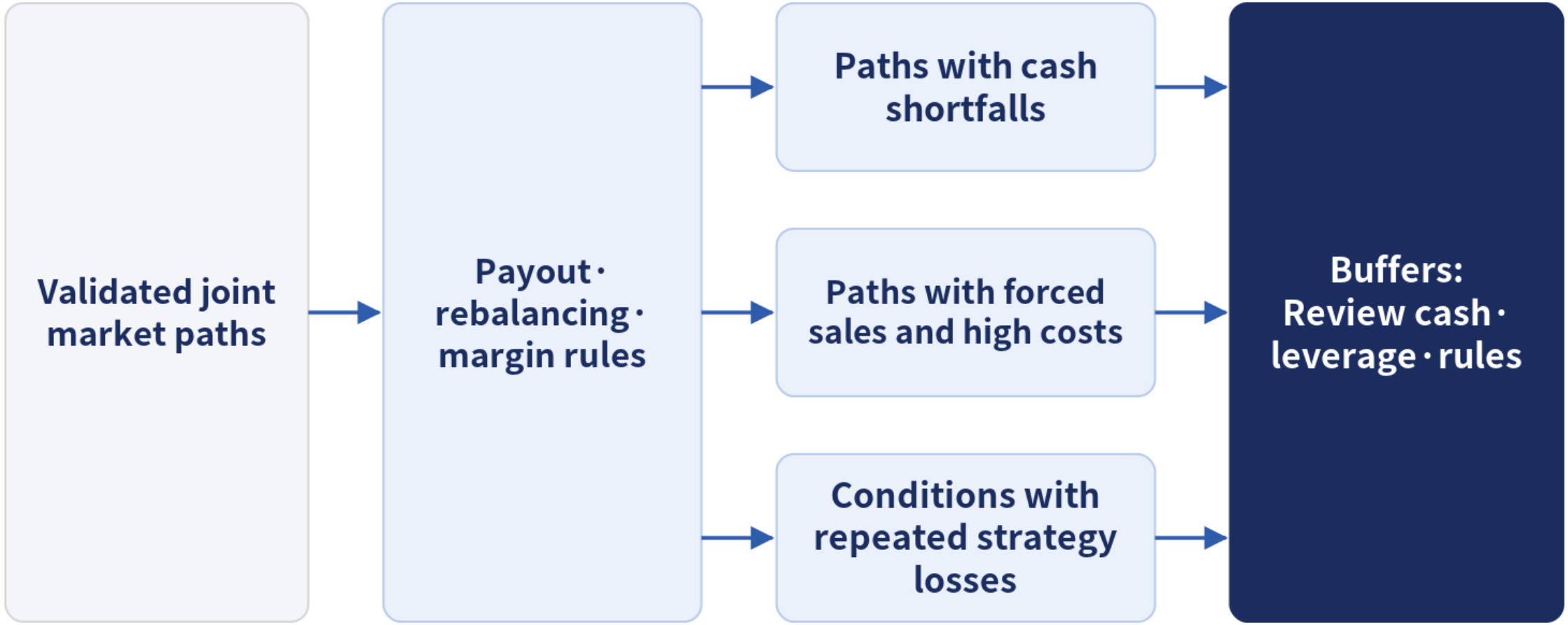


Overlapping Evaluation Windows

When 20-day outcomes overlap, evaluation samples are dependent — also evaluate on non-overlapping origins

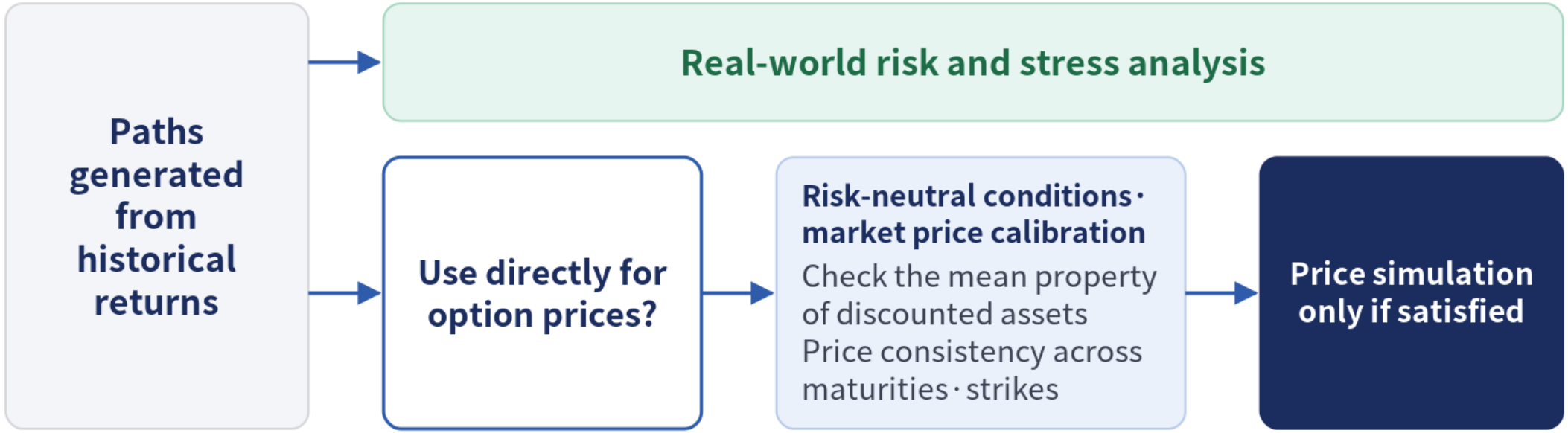
Apply payout and margin rules to joint paths

If cash flows themselves are random, generate their relation to the market jointly or state a separate assumption



Option prices go through a separate door

The discounted mean payoff of a historical-path GAN is not automatically an arbitrage-free option price



Risk-neutral construction in Quant GAN

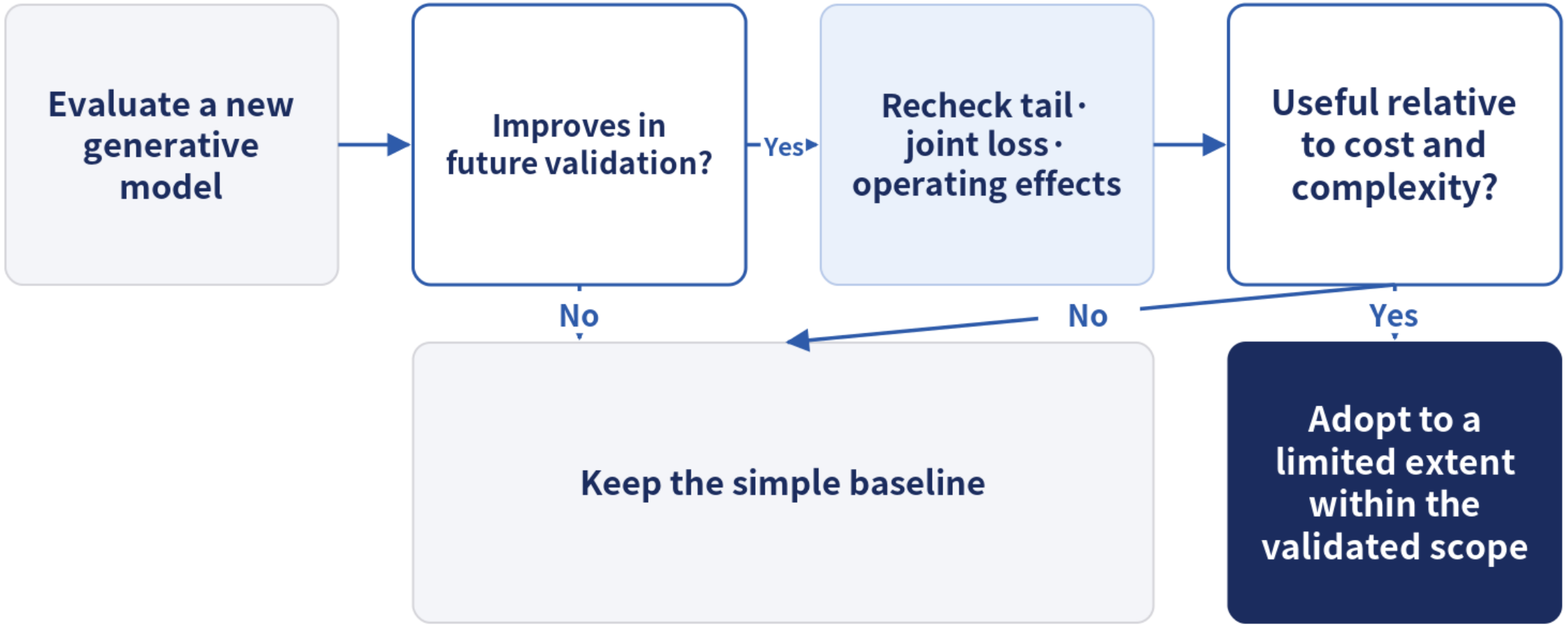
A property of a specific construction; do not extend it to all GANs

Scope

Real-world risk simulation is the core; option price generation is an advanced topic

Adopt the enrichment only within the validated scope

If a complex model loses to a simple baseline, that result is itself a valid outcome



An enrichment verdict report has five components

Include failure conditions in the conclusion

Generator Structure

Structure diagram of traditional, bootstrap, and generative

Distribution & Dependence

Real vs. synthetic distributions, clustering, joint-decline plots

Loss Distribution Comparison

Losses and cash shortfalls under the same operating conditions

Out-of-Time Risk Forecasting

Frequency and clustering of exceedances in real future windows

Failure Conditions

Under which conditions did it not help?

Limit the conclusion to where generative models help more than a simple baseline

04

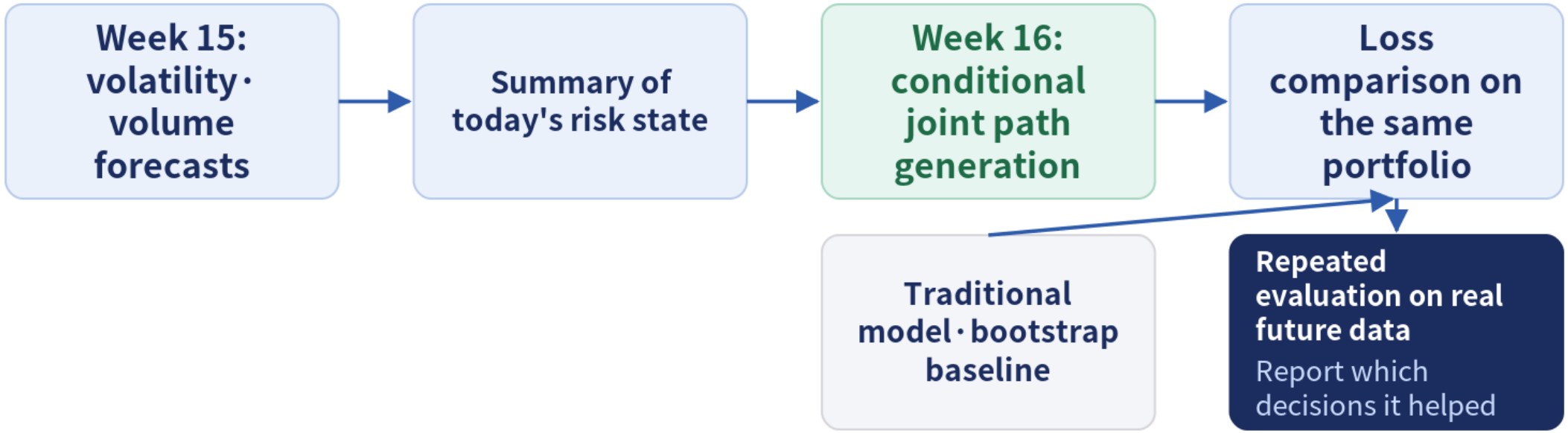
PART 4 · Capstone

Linking Forecasting and Simulation Once

Week 15 forecasts → Week 16 conditional generation → same portfolio evaluation

Forecasts supply conditions; generative models build paths

A teaching system design; it does not mean the original papers validated this whole combination



Quantiles ≠ Joint Distribution

Do not treat stitched quantiles of a pretrained forecaster as the generator's joint distribution

Error in Conditions

Using forecasts as conditions carries their forecast errors along

Enter result numbers only after running

Compare the simple baseline with deep-learning and generative candidates on the same future window and rules

Item	Judgment Question	Simple Baseline	Candidate
Forecast Accuracy	Improved on the same future window?	After run	After run
Forecast Interval	Did the stated range actually cover?	After run	After run
Loss Tail	Did it avoid missing extreme losses?	After run	After run
Joint Decline	Does it avoid overstating diversification?	After run	After run
Risk Forecast Exceedances	Are exceedances concentrated in certain periods?	After run	After run
Payout · Margin Failure	Does fragility show under the same rules?	After run	After run
Cost · Time	Does it justify the added complexity?	After run	After run

Lab 2 is an introduction to portfolio Monte Carlo

Join the two boxes and paste them at once · Check: consistency of co-movement and payout/cost accounting

[Input] Prompt · Claude Code · ① Setup

```
Write a picture-based, equation-free
Monte Carlo introduction lab.
Initial assets are KRW 10bn; target weights
are equities 50 · bonds 30 · gold 20.
Pay out KRW 50m monthly, rebalance
quarterly, and assume a 10bp
transaction cost.
Specify the initial cash balance, the
selling order for payouts, and the
basis for charging transaction costs.
Also fix first how shortfalls in
payouts are recorded.
```

[Input] Prompt · Claude Code · ② Generation and Output

```
First, generate synthetic market data
for teaching and label it as assumed.
Baseline generators: a multi-asset normal
return model and a same-date block bootstrap.
Generate one-year joint paths while
preserving correlations.
Compare 1,000 vs. 10,000 paths and several seeds.

Output terminal wealth, 95% VaR and ES of terminal loss,
maximum drawdown, payout failure rate, and
total transaction costs as tables and plots.
Explain the loss sign and horizon of VaR and ES.
Distinguish differences between models from
differences across random repetitions.
Do not present shares of generated paths as
confirmed real-market probabilities.
```

Lab 3 trains one GAN or Diffusion model

Check: whether it actually trained · conditional vs. unconditional · independent test window

[Input] Prompt · Claude Code · ① Design and Training

```
Design and run an experiment that
enriches the baseline Monte Carlo generator
with a generative time-series model.
Check the compute budget first. If GPU is
short, use a small model and short windows.
Do not claim performance is validated
on the grounds of reduced training.
Build multivariate windows from
same-date returns of several assets.
Fix training, validation, and test dates first
and remove boundary overlaps.
Actually train one GAN or Diffusion
model and record its version and seed.
If possible, condition on recent market history
and jointly generate the next 20 days.
If conditional implementation is impossible, state
that generation is unconditional, and do not claim
conditional risk-forecasting performance as of today.
```

[Input] Prompt · Claude Code · ② Comparison and Reporting

```
Compare the traditional model, block bootstrap,
and learned generative model under the same conditions.
Plot distributions, crash tails, volatility clustering,
and joint losses in downturns.
Also check whether synthetic paths
copied training paths.
Do not use the real future window for model
selection; use it only for the final evaluation.
Apply the same operating rules to the 20-day paths.
Do not claim a full year is validated by
arbitrarily stitching 20-day windows together.
Report training instability, missed rare crises,
compute cost, and evaluation failures as they are.
Limit the conclusion to the conditions under which
generative models help more than the simple baseline.
Keep equations out of the result document;
explain with structure diagrams, validation plots, and
tables.
```

Lab 4 shows the principles through visualization

A lab that explains how GANs and Diffusion work without training a generative model

[Input] Prompt · Claude Code

```
Build a visualization lab that shows
GAN and Diffusion principles to non-developers.
Do not call samples made with normal draws or
the bootstrap GAN or Diffusion results.
If a trained model cannot be run, label
it as an animation explaining the structure.

For GANs, show the discriminator separating real and
generated paths and the generator's correction process.
For Diffusion, distinguish the training-side example
of adding noise to an original path from the
generation-side process that starts from new noise and removes it.
Do not substitute images from image generation
for financial time-series samples.
Clearly separate model run results from
schematic explanations on screen.
```

What to Check

- Are run results and explanatory schematics separated on screen?
- Are the training and generation directions distinguished?
- Were fake samples not called model results?

Core Literature on Generative Time Series

Read general time-series results, financial generation experiments, and applied designs separately

No.	Source	Year · Status	Scope in this lecture
S5	TimeGAN	NeurIPS 2019	GAN accounting for temporal dynamics
S6	Quant GANs	Quantitative Finance 2020	Financial time series · TCN · specific risk-neutral construction
S7	Diffusion-TS	ICLR 2024	General time-series generation architecture
S8	TimeGrad	ICML 2021	Conditional multivariate probabilistic forecasting
S9	GAN–Diffusion hybrid for financial time series	2026-05 preprint	Recent experiment reflecting correlation structure

arXiv 1907.06673 (S6) · 2403.01742 (S7) · 2605.27113 (S9) · PMLR v139 (S8) · NeurIPS 8789 (S5)

Generated paths matter when judged by the same yardstick

The final lecture of the 16-week course — tying forecasting and simulation into one decision flow

1

Monte Carlo = repeated generation + same rules + summary

Look not only at the mean but also at the lower tail, maximum drawdown, and cash depletion frequency.

2

Generative models enrich joint paths and shocks

Shock learning · joint generation · scenarios by condition — beware double counting and date separation.

3

The verdict is made in the real future

Adopt only to a limited extent when out-of-time risk forecasting and operating effects justify cost and complexity.

A complex model losing is also a valid learning outcome